

AI/ML Software & Integration: Army IVAS, Integrated Visual Augmentation System

Capability Overview

The Impact of Artificial Intelligence on Military Operations

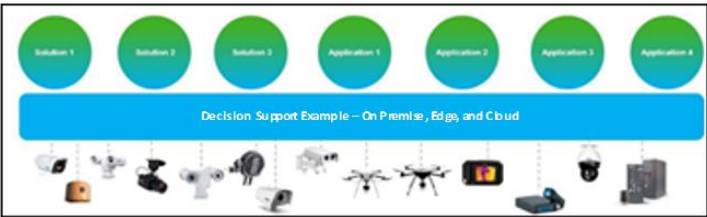
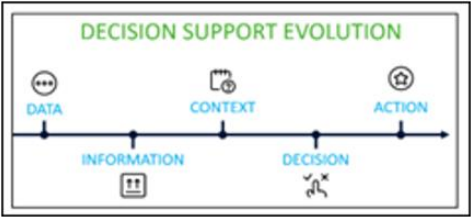
As the world becomes increasingly reliant on technology, the field of artificial intelligence (AI) is becoming more important in areas such as machine learning (ML) and computer vision (CV). Its applications in unmanned ground vehicles (UGVs) and unmanned aerial systems (UAS) have driven substantial growth in recent years, with the global market projected to expand at a compound annual growth rate of 35% over the next five years. This growth has been driven by the U.S. Department of Defense's (DoD) search for asymmetric advantages to enhance mission readiness and effectiveness, such as unmanned systems, autonomous capabilities (AI Agent Behavior), and teaming capabilities (HMT+MuM-T). As the DoD attempts to address the issues of army recruiting shortfalls and USAF pilot shortages, AI technology has the potential to revolutionize the way the military operates, by providing greater situational awareness (SA) and reducing risk to mission while simultaneously increasing **DECISION SUPPORT** and **EFFECT** speeds, where:

DECISION SUPPORT	EFFECT
- Unmanned scene & threat Assessment - Human Situational Awareness, SA - Notification to nearby team members - Engagement approval from JADC2	- Electronic Warfare + Jamming - Use of Force – Human Engagement - Use of Force – Unmanned + Manned Engagements - Kinetic Strike



eXeL AI Value Prop: Enhance Battlefield Understanding and Reduced Risk with Multi-Domain Operations

Multi-domain operations necessitate a teaming of **Field Operatives**, **Officers**, and **Joint Force Commanders** leveraging various levels of technology, such as soldier-worn cameras (body worn + IVAS), unmanned ground vehicles (UGV), small unmanned aerial systems (sUAS) and medium unmanned aerial systems (mUAS). Additionally, larger aircrafts like the fly-by-wire F16 and unmanned Fury's must integrate and aid in creating a common operating picture through radio and Link16/22 systems, combining human-operated sensor suites with those of UGV and UAS to facilitate real-time geo-location of targets of interest, thus providing an enhanced battlefield understanding. **Field Operatives** benefit from a reduction in cognitive load with edge AI capabilities enabling real-time decision making, risk assessments, and action appropriate to the situation. **Officers** are given trusted real-time data for situational awareness and decision making, while **Joint Force Commanders** benefit from precision execution of operations leading to a decrease in risk to mission, force, and defense assets. An eXeL AI enabled IVAS system, which expands its capabilities through sensor integration, computation and AI/ML based abilities, facilitates training, pre-mission planning, mission execution, and post mission debriefs, with a digital playback of the mission providing a comprehensive analysis of the team's performance.



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eXeL AI Platform Benefits: Enhanced Autonomy, Improved Decision Making, Flexibility and Scalability

The eXeL AI Platform Ecosystem provides an integrated solution of IVAS and interchangeable manned and unmanned systems, leveraging remote imaging sensors, artificial intelligence (AI) + computer vision (CV) models, and edge computing to autonomously execute user-defined tasks. This platform is equipped with identity (ID), geo-location, and fusion capabilities, allowing users to access mission specific AI and combine AI/ML algorithms from UGVs, UASs, and aircraft video feeds accessed through IVAS and an either Mimic or Universal Controller. Additionally, it includes modular edge compute capabilities which enable the recording and playback of real-world missions via a digital twin, facilitating AI autopilot adaptive execution and integration into Joint All Domain Command and Control (JADC2) control stations. This system has the potential to provide higher confidence decision-making and risk reduction in a variety of scenarios, such as public safety, aerial operations, and high risk defense operations, as well as allowing for expansion over time with additional hardware, sensors, AI/ML models, and autonomous unmanned flight and robotic behaviors.

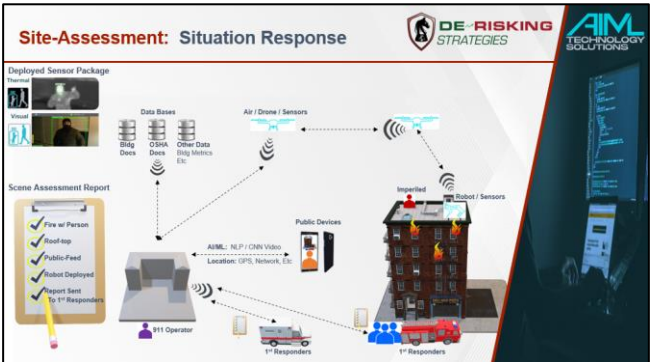
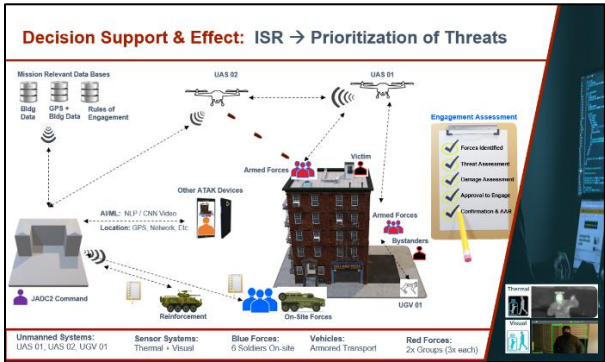
eXeL AI Platform: Prior Version



The Value of Reducing Operational Metrics: eXeL AI

Primary operational metrics for focus in the Defense Department's BlueFor operations include **Risk Assessment Quality**, **Speed to Decision** and **Speed to Effect**. Secondary metrics are concerned with the **Resourcing** of agents, risk analysts, decision support personnel, and approvers/commanders. Tertiary metrics are utilized to **Optimize** the value of physical assets at risk, the number of data hand-offs, bandwidth limitations, and total data transferred. By reducing key metrics, partnered organizations can leverage pay-for-performance and value-share models to extract ongoing value from complex DoD engagements that include IVAS and synergistic deployments. Each tactical or defense operation has the potential to incur losses of up to two to five million dollars, not accounting for potential loss of life, casualties, and asset loss. This can amount to a total annual expenditure of between \$100-150 million dollars, depending on the complexity and number of operations for a given unit.

Tactical + Non-Tactical Visual: CONOP Overview



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The Team:

Alex Seguin has extensive experience in global business management, having held positions at Honeywell, Fortive, and FLIR, where he was responsible for launching new products and guiding R&D portfolio prioritization for executive management. Most recently, he worked at Athena AI, which leverages multi-layer machine learning models, where he drove US strategy and AUSA progress resulting in successful partnerships with RedCat, Teal, and Tomahawk. Notably, he launched the [ii900 product](#), an acoustic imager, which achieved \$20M in revenues during its first year, four times greater than the proforma volume. At FLIR, Seguin worked directly with the CTO team to prioritize a \$200M R&D budget in preparation for an acquisition by Teledyne, and for the integration of the Muve 360 sensor package from Boston Dynamics. Seguin holds a chemical engineering degree from the University of Texas (UT-Austin) and an MBA from Auburn University. Additionally, he has furthered his knowledge in the area of AI, having completed a 6-month post-graduate program at UT-Austin's McCombs School of Business for AI/ML machine learning, and has completed an 18-month Harvard Business Analytics Program.

Alex Seguin - Principal Investigator

Institution	Location	Major/Area of Study	Degree	Year
- University of Texas	- Austin, TX	- Chemical Engineering	- BSCE	- 2008
- Auburn University	- Auburn, AL	- Masters in Business Administration	- MBA	- 2016
- MIT	- Boston, MA	- AI Implications to Business Strategy	- Certificate	- 2018
- University of Texas	- Austin, TX	- PGP AI/ML - Business Analytics	- 6-mth Cert	- 2021
- University of Harvard	- Cambridge, MA	- Business Analytics Program	- 18-mth Cert	- Jan 2020-Mar 2022

Technical Solution: Integrating Artificial Intelligence Platforms for Multi-Domain Operations

This technical proposal outlines the objectives of a project to develop an AI platform ecosystem for multi-domain operations, with the aim of enhancing the integration and value of the IVAS. The project will involve the integration of unmanned ground vehicles (UGVs), unmanned aerial systems (UAS), and custom drones with eXeL AI's real-time geo-location + GPS of targets of interest, edge neural net deployments, and ruggedized deployment of additional mUAS aircrafts. To achieve these objectives, the project will require expertise in AI/ML training, sensor technology, data annotation and labeling, data augmentation, auto labeling, software architecture, modular compute options for different size, weight and power (SWaP) needs, hardware-in-the-loop simulation and AI agents and behaviors for accelerating mission planning.

Phase 1 of the project will involve integrating three drones into a multi-drone mission with real-time output into any IVAS system on the network. Phase 2 will involve integrating a UGV, a heavy lift drone with laser designation, and a custom drone equipped with an Axon taser, all equipped with eXeL AI's edge processing and equipped with "In App" escalations and approvals. Phase 3 will involve ruggedized deployment of additional mUAS aircrafts, such as AeroVironment and Quantum Systems, to complete a high priority CONOPs with basic JADC2 integration, for Project Convergence '23/'24. These aircrafts will be equipped with basic autonomous AI flight behaviors. Finally, phase 4 of the mission planning process will involve the implementation of Honeywell SatCom, a VBAT 228 and Trillium AI edge capable payload (HD95-MVS-LD) to enable remote operations for fixed-wing laser designation.

The project will require a team of experts with experience in AI/ML training, sensor expertise, data annotation and labeling, and software architecture. In order to succeed, the project will also require modular compute options for different SWaP needs, as well as hardware-in-the-loop simulation and AI agents and behaviors for mission planning. The successful completion of this project will provide real-time insights to government and enterprise organizations through IVAS operations & missions, reducing cognitive load to operators, and allowing for fit-for-purpose AI/ML packages to be downloaded and applied to each sensor.

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Technical Objectives:

1. Identification and Significance of the Problem or Opportunity.

The U.S. Department of Defense (DoD) is looking for ways to gain an advantage over their adversaries, and artificial intelligence (AI) technology is being developed to help them do so. The eXeL AI Platform Ecosystem evolves via AI-powered and capable systems that integrates IVAS with mobile sensor systems, fixed imaging sensors, and edge computing. It can give real-time insights to government and enterprise organizations, reduce cognitive load to operators, and provide a common operating picture for multi-domain operations. This platform enables faster, more accurate decision-making, and can be used with soldier-mounted cameras, IVAS, UGVs, sUAS, mUAS, and fly-by-wire F16s. It also provides enhanced situational awareness, real-time decision making, risk assessments, and a digital twin that can use AI autopilots for adaptive execution. On top of this, it can be easily expanded with additional hardware, sensors, AI/ML models, and autonomous AI flight behaviors. Overall, this platform has the potential to revolutionize the way the military operates, and the DoD is investing in the development of AI technology to gain an asymmetric advantage. It can help operators make faster, better decisions, and provide a fit-for-purpose AI/ML package that can be downloaded and applied to each sensor.

2. Key Technical Challenges and Risks

Risk is an important factor to consider when it comes to the deployment of AI and ML capabilities for military use. Expertise in AI/ML research, DevOps, and Consultancy is required to ensure optimal models are executed properly. Modular compute options need to be available to scale up and down with different TOPS and fit-for-purpose SWaP needs. Additionally, hardware-in-the-loop simulation and AI agents and behaviors are essential to reduce mission risk. Furthermore, the integration of an 850 nm infrared laser and a LiDAR with an electromagnetic signature is an effective indicator for hostile forces. To achieve real-time sensor fusion and geolocation, large advanced compute packages, such as Palantir's TITAN, are required. However, these packages have yet to be widely deployed. To successfully deploy AI and ML capabilities, risk management measures need to be taken and the appropriate expertise and tools must be employed.

3. Phase 1 - 4 Work Plan Outline

Scope: Human-machine teaming with UGVs, medium UAS, AI edge compute, and SatCom tied into IVAS, ATAK, and Kågwerks, for executing missions with prioritized real-time outputs to aid in SA, escalations, and approvals.

Technical Objectives By Phase

1. *Integrating Drones into a Multi-Drone Mission with Real-Time Output into real-time 3D Engine*

Phase 1 incorporates 3 drones (FLIR, Teal, etc) to create and complete a multi-drone mission with integration into real-time output into a Live2Sim demo (Cesium's Unreal 5 + One World Terrain, or equivalent).

2. *Integrating UGV, Heavy Lift Drone, and Custom Drone with eXeL AI Edge Processing*

Phase 2 integrates a UGV (Ghost Robotics, Boston Dynamics, ect) along with a heavy lift drone with laser designation and a custom drone equipped with Axon taser, all equipped with eXeL AI's edge processing into Tomahawk's RAID and/or Kinesis platform which already integrates into Athena AI detection and ATAK.

3. *Ruggedized AI Deployment of mUAS Aircrafts + Project Convergence '23/24:*

Phase 3 includes Project Convergence '23/24 and will include JADC2 integration and ruggedized deployment of additional mUAS aircrafts, such as AeroVironment and Quantum Systems, with basic autonomous AI flight behaviors from partners to complete high priority CONOPs with UGVs and sUAS.

4. *Mission Planning Process: Utilizing SATCO, VBAT 228 and Trillium AI Edge Payload*

Phase 4 of the mission planning process involves the implementation of Honeywell SatCom, a VBAT 228 and Trillium AI edge capable payload (HD95 – MVS - LD) to enable laser designation on a fixed wing VTOL unit. This will provide simulation of various desired effects.

Task Outline:

- ✓ Ph1-TO-01: MVP of 3x drones with existing FLIR sensors and Muve 360 MVP1.0 w/ IVAS integration
- ✓ Ph1-TO-02: Software MVP for UI/UX with drafted UI/UX of auto-notifications highlighting targets of interest
- ✓ Ph2-TO-03: Integration of single GPU for AI edge compute along w/ UGV integration
- Ph2-TO-04: Integration of multiple AI edge compute/GPU modules + heavy lift LD UAS and custom taser drone
- Ph2-TO-05: IVAS integration to Tomahawk controller w/ AI edge compute, Kågwerks & In-App approvals
- Ph2-TO-06: Software Design with auto-notification & "In-App" real-time escalations and approvals

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- Ph3-TO-07: Testing 01 of Alpha eXeL AI MVP1.5 with IVAS accessible approvals
- Ph3-TO-08: AI/ML Model Dev 01: Iterate on selected AI/ML models, AI autonomous behaviors, and use cases
- Ph3-TO-09: Final hardware selection and production ready enclosures and deployments
- Ph3-TO-10: Testing 02 of Beta eXeL AI Prototype with an MVP2.0 of edge compute and integration for JADC2
- Ph4-TO-11: AI/ML Model Dev 02: Securely LIVE update new AI/ML models & autonomous AI behaviors
- Ph4-TO-12: SatCom integration of Honeywell system, and A2AD demo of MVP2.5
- Ph4-TO-13: Final Hardware and Software deployment with most recent Software and AI models (MVP3.0)

Description

Ph 1: Drone Mapping Integration
Ph 2: Robotic Automation Integration
Ph 3: UAS/UGV AI Deployment
Ph 4: Testing / Prototype Iterations

Milestone Schedule

Kickoff + 6 months
Kickoff + 9 months
Kickoff + 12 months
Kickoff + 18 months

Deliverable

MVP Demo + Deployed UI/UX w/ Approvals
Robotic Automation Integration
Artificial Intelligence Deployment
Mission Planning Optimization

Progress Reports

- Planned report outs after each desired milestone, to be outlined in SOW (statement of work).
- Alpha Unit feedback on system MVP1.0 designed for customer feedback.
- Beta Unit feedback on system MVP1.5 designed for prototype iterations
- Delivery of MVP2.0 with specific feature enhancements outlined for commercial ready system (MVP3.0).
- Launch commercial ready system of UGV/UAS tied to IVAS system, with decision making protocols built-in.

ROM: Rough order of Magnitude

Phase 1- 4 Total Spend: \$18,000,000

	Ph 1: Drone Mapping Integration	Ph 2: Robotic Automation Integration	Ph 3: UAS/UGV AI Deployment	Ph 3: Testing / Prototype Iterations
Labor # FTE	5	10	15	20
Labor Costs	\$1,000,000	\$2,000,000	\$1,500,000	\$2,000,000
Material Asset - Costs	\$200,000	700,000	1,500,000	1,600,000
Partnership + Investment	\$2,000,000	\$2,000,000	\$2,000,000	\$2,000,000
Total, \$	\$2,200,000	\$4,700,000	\$5,000,000	\$5,600,000

Broader Impacts:

1. Increasing the **economic competitiveness** of the United States.
2. Advancing of the **health and welfare** of the American public.
3. Supporting the **national defense** of the United States.
4. Enhancing **partnerships** between academia and industry in the United States.
5. Developing an **American STEM workforce** that is globally competitive through improved pre-kindergarten through grade 12 STEM education and teacher development, and improved undergraduate STEM education and instruction.
6. Improving **public scientific literacy and engagement** with science and tech in the United States.
7. Expanding participation of **women, minorities, and veterans in STEM and AI**

Preparing AI to Serve the Future of Humanity

The potential of the eXeL AI Platform Ecosystem to revolutionize the military is tremendous. With the ability to provide real-time insights, reduce cognitive load for operators, and incorporate digital twin technology for adaptive AI autopilot execution, the IVAS enabled platform could drastically improve the efficiency and efficacy of government and enterprise organizations. It could also allow for fit-for-purpose AI/ML packages to be downloaded and applied to each sensor, enabling more efficient operations. In the future, the platform could lead to a new era of automated operations, with AI systems performing complex tasks, providing enhanced situational awareness and real-time decision support with minimal human input. With these advancements, the eXeL AI Platform Ecosystem is poised to become a leader in the field of artificial intelligence and its applications.

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Appendix

Market Landscape & Relevant Programs of Record: Unmanned UAS + UGV

Application Alignment	Value Creation Opportunities
Autonomy: Manned w/ AI, Remote control w/ AI, Unmanned - fully autonomous AI Wingman: AI co-pilot, AI enabled training, Unmanned AI aircraft protection Intelligence: Remote piloted aircraft with AI intelligence / ISR optimized capabilities Defense Logistics: Manned/Unmanned AI delivery, AI inspection for maintenance Interstellar: Space/Mars specific AI powered aerial craft. Note: Far future Other: Air Launch Effects (ALE) + Loitering Munitions Aircraft Integration Possibilities Multi-sized aircraft Group 1/2/3: Quad/VTOL/Fixed, Group 4/5 integration partners, Larger Quads: Electric/Fuel/Jet, Helicopters, Ground Systems + Sea Vessels. Programs of Record + Trade Shows Space / Air Force Teaming: Golden Horde, Skyborg Other: NTS-3, Rocket Cargo Army Programs Mobility: FIUAS/TUAS/EUAS, FVL CFT, NGCV (RCV, OMT), CRS-H, CRS-H, AMPV, AWS Decision: AI2C, ITAS, EASIT, Vantage, Cyber SU, SL CFT, GFT, ARL-M, APNT, N-CFT, Air SS, ITN Navy/Marines: M12, ONR, AWS-UAS SOF: PEO-C4, PEO-FW, PEO SDA DHS: C-UAS Trade Shows Army: AUSA Air Force: AFA Navy: Sea, Air Space Marine Corps: Modern Day Marine international: DESI + Eurosatory Non-Service Specific, Industrial OR commercial: AUVSI-Xponential, AUVSI-DPS, CES. Publications: Janes, Shephard, National Defense Magazine, Defense Daily, Breaking Defense	Enhancements AI-Payloads: Modular sensor add-ons, integration speed, upgradable AI for vision Modular Compute: Ability to quickly replace & scale AI compute: Edge TPU/GPU/etc. Propulsion Options: Gas, Heavy/Jet Fuel, Hydrogen, Batteries/Hybrid, Ion Thrusters Custom AI Missions: Auto-data recordings, user trained AI, user optimized missions Field AI Upgrades: Cyber-secure upgrades for new AI model testing + management Charging Options: Low maintenance charging stations, Auto-battery replacements Targeting Capabilities: Human + AI on moving targets w/ casualty + RISK assessment Stealth Options: Stealth fixed/VTOL aircraft integration partners Integrations Teaming: Multi-UAS/UGV missions, AI enhanced insight, & Human-in-the-Loop [HITL] AI Ranked Threats: Prioritized threats w/ Fixed, PTZ, & GRD/AIR mobile AI systems AI Tracking / Assist: Auto-tracking – separate control of payload system HITL Decision: AI assessed threats, actionable insight, enhanced HITL decision speed Extra Edge Compute: Off-aircraft AI edge compute or mobile Edge drone Remote Console: Command/control center, cyber-secure platform APIs, dual-view AR/VR: Display integrations, with optimized AI/ML with controls via voice/touch/etc. Deployment: Ground: Human: Manual/Semi-auto deploy, Auto-deploy: stationary + vehicles + box Aerial: Aerial deployment, Booster/rocket aerial assist Water: Ocean/Lake vessel, Sub-sea to aerial Recovery Ground: Human, Ground mobility system: Remote + Autonomous AI recovery Aerial: Pickup Day Drone, Pickup Night Drone Damaged: Field destroyed field unit: SW/AI memory wipe + deploy HW interlocks.

PDF + Videos:



DoD AI Thoughts



UAS Teaming + XR Thoughts



SBIR DHS Brief



SBIR DHS Tech Proposal



IVAS: Future Infantryman



Sample: Tech Volume

Inspired By:

Events	Videos	Books & Magazines	
ITSEC 2021 AUSA 2022 IAPPA 2022 AF Weapons Pitch Day 2022	TESLA AI Day 2021 TESLA AI Day 2020 TESLA Car Owner* DARPA Dogfight Trials Cesium Unreal 5	Army of None Kill Chain Future of War Army - AUSA National Defense	A Thousand Brains Competing in the Age of AI AI Super Powers AI 2041

Companies	Products
Palantir Anduril Shield AI FLIR Fluke	Trillium Engineering RedCat (Teal/Tomahawk) Reveal Axen Honeywell Shield AIVBAT Shield AI Nova Tomahawk Kinesis FLIR Hadron 640R Honeywell SatCom Fluke SV600 Fluke ii900/910 eXeL AI Educational Kit

Contacts and References:

Deloitte: Leo Hmelnitsky, Robert Duffy
Professional: Mark Holden, Clint Wiechert, Paul-Devin Kuettner